Assignment #1 Technology Infrastructure: Networking INFO1380 – Fall 2026

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Course: Technology Infrastructure: Networking

INFO 1380 – Fall 2026

Instructor: Todd Antoine

Due: September 29, 2026

Network Topology

You have been contracted by a local manufacturing company, Maccrage's Foundry, who has made the request for you to build for them a simple network. When meeting with the president of the company, Robert Gullman, he has indicated that he wants his network to be able to support the devices currently deployed, as well as considering the network will likely expand. He has a vision for his company and wants to know what it is that you plan to do for him as a contracted networking technician.

Submission Instructions:

- Submit the assignment in one PDF file and one PKT file ONLY before the due date.
- Ensure that all screenshots are attached to the PDF file as part of your submission.
- Failure to submit as a PDF document, or having screenshots separate from your PDF file will result in the refusal of marking this assignment.
- **Failure to submit the assignment by the documented due date will result in score deduction.**
- **Failure to submit the screen shots or incomplete screenshots (ones without description) will result in score deduction.**
- Ensure that all configurations are properly labelled, showing Hostname, IP addresses, and directional flow of traffic

Assignment Objectives:

You will build a simple network that includes end points, intermediate devices, network hardware, and printers. This network needs to be properly mapped, clearly thought out, designed, labelled, and presented to the client.

TOPOLOGY INSTRUCTIONS: (5 points)

(a) Using Packet Tracer, build a star topology that includes the following items:

a. 1 4331 router

b. 1 2960 switch

c. 4 endpoints (either laptop or desktop)

d. 2 printers

(b) **INCLUDE A SCREENSHOT OF YOUR TOPOLOGY. ENSURE IT INCLUDES MEANINGFUL HOSTNAMES, IP ADDRESSES, AND DIRECTIONAL FLOW OF TRAFFIC.**

NAMING CONVENTIONS: (5 points)

All devices on your network will need to be properly named to have meaningful names associated to the entire network.

(c) Your network should be found on the **192.168.1.x** network, and all devices should be assigned an appropriate IP address from this network.

a. Your endpoints should be assigned an IP address between the range of **192.168.1.10 to 192.168.1.20. SHOW A SCREENSHOT OF THIS SETUP.**

b. Your printers should be assigned an IP address between the range of **192.168.1.21 to 192.168.1.25. SHOW A SCREENSHOT OF THIS SETUP.**

(d) Endpoints should be named as **[STUDENTNAME]WS##** or as **[STUDENTNAME]LT##**. This will depend on whether the device is a desktop or laptop. **SHOW A SCREENSHOT OF THIS SETUP.**

(e) Your printers should be named as **PRNT##**. **SHOW A SCREENSHOT OF THIS SETUP.**

CLI commands: (5 points)

a. Your router should be renamed as **ROUTER-MAIN** for the hostname, while the Display Name of the device should be meaningful to the device and its function. The renaming of the hostname should be completed in the CLI. **SHOW A SCREENSHOT OF THIS SETUP.**

b. Your router should use the IP address of **192.168.1.1**, and this should be executed through the CLI. **SHOW A SCREENSHOT OF THIS SETUP.**

c. Your switch should be renamed as **SWITCH-MAIN** for the hostname, while the Display Name of the device should be meaningful to the device and its function. The renaming of the hostname should be completed in the CLI. **SHOW A SCREENSHOT OF THIS SETUP.**

Theory: (10 points)

(f) **Maccrage's Foundry is pleased with the network you have designed. The company expects to add additional computers and printers over the next few years. Based on the network you created for this assignment, what changes or improvements would you recommend to ensure the network can support future growth?**

I recommend the introduction of hybrid topology. Because star topology was used, it would be smart to integrate another topology type to ensure flexibility, though some rerouting might need to be done. And since the Foundry is debating upscaling, I recommend that they use DHCP since they are adding more endpoints. DCHP also helps in making the setup of endpoints easier. Finally, I recommend using CIDR to set however many IP Addresses to company needs to represent the network. It not only allows for more efficient assigning of IP addresses but also does not waste any unused addresses.

In your answer, discuss:

a. IP address planning

b. Network hardware requirements

Provide at least **three recommendations** and explain how each recommendation would improve the network's scalability, reliability, or manageability.

(g) Maccrage's Foundry is pleased with the network you have designed. After reviewing your proposal, Robert Gullman asks what recommendations you would make to improve the network's performance, security, reliability, and ability to support future growth. Identify and explain at least three improvements you would recommend for this network. For each recommendation:

a. Describe the improvement.

b. Explain why it would benefit the organization.

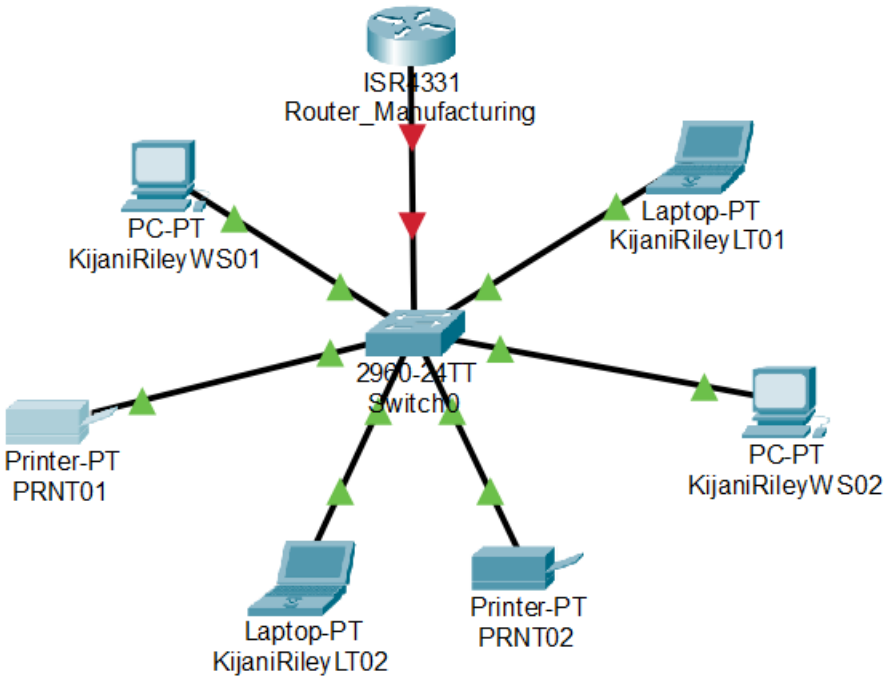
c. Explain how it would support future business growth, security, or reliability.

Your answer should be written in complete sentences and demonstrate your understanding of networking concepts.

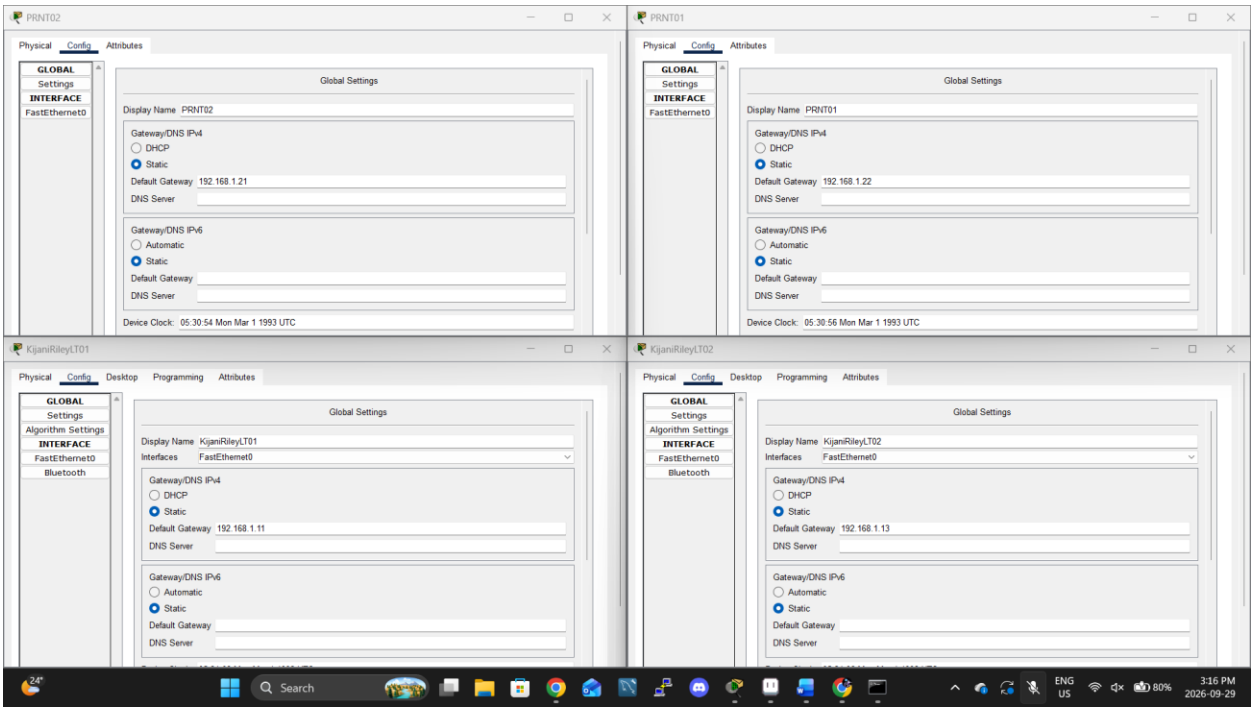
Firstly, I believe that CIDR should be taken advantage of. CIDR allows for the breaking up of IP addresses into smaller networks, which allows an organization to assign what addresses they need and not have to worry about any unused addresses. It benefits the organization, not only by reducing traffic noise and enhancing security, but giving networks an easier time to grow.

Secondly, private IP addresses should be used if they aren't already. Since most, if not all endpoints, are in the building where the network is, it would enhance security because private IPs are not visible online. A VPN could also be an addition in case users need to work from home.

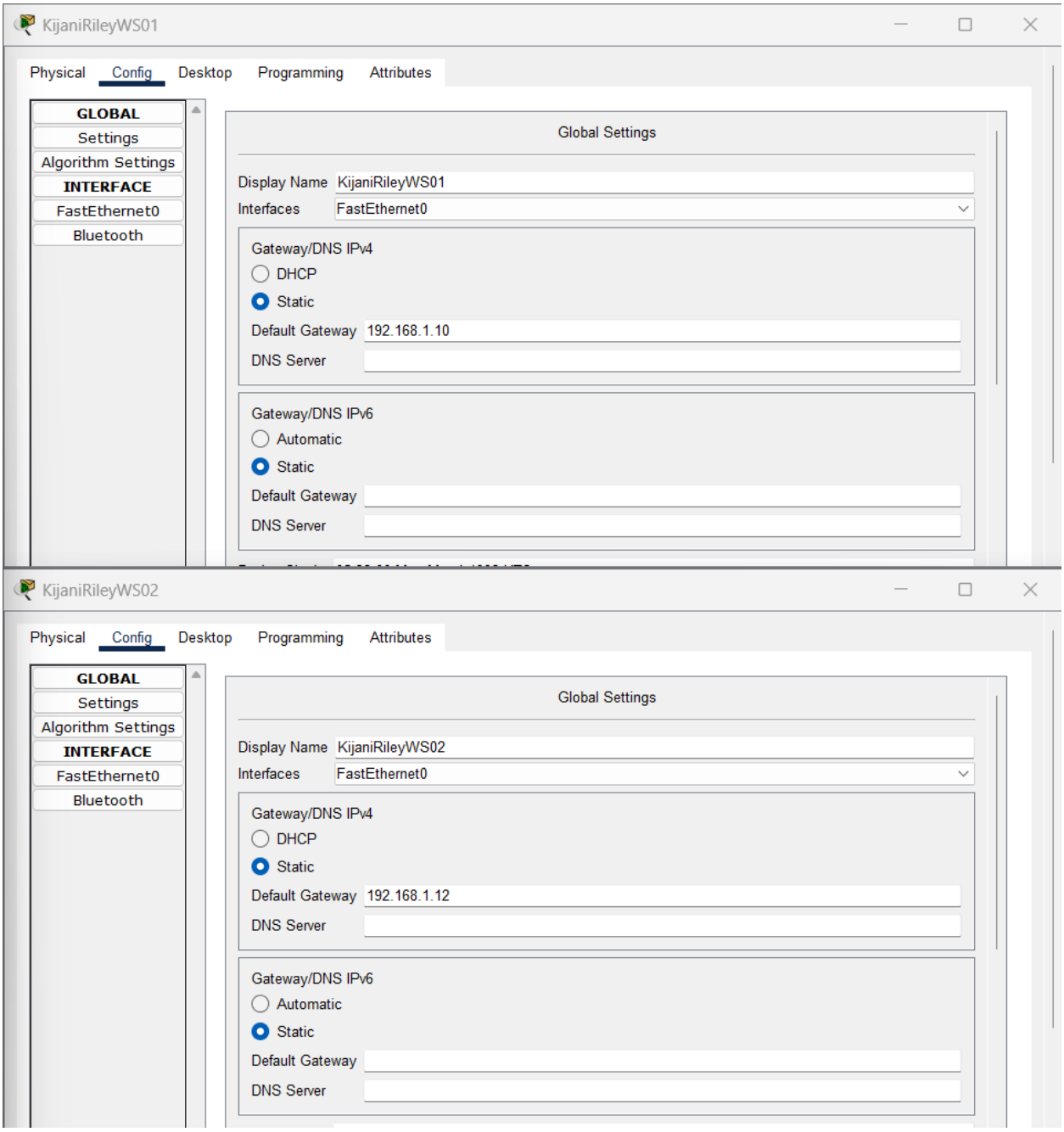
Lastly, the renumbering method should be used during the network redesign. Renumbering is the idea of changing IP addresses used on a network. This method works well when redesigning networks and scales well, meaning that it can fit the needs of the organization, however dynamic. If they need to dial back on endpoints sometime down the future, renumbering, allows you to assign addresses to some devices, while CIDR removes the issue of any extra addresses by breaking the down.



- The star topology network, consisting of 6 endpoints, a 2690 switch, and a 4331 router.
  (2 Laptops, 2 PCs, and 2 Printers).



- Setup 1 of 2: IP Address assignment to KijanirileyLT01, KijanirileyLT02, PRNT01, and PRNT02.
- These endpoints were assigned these IP Addresses: 192.168.1.11, 192.168.1.13, 192.168.1.22, and 192.168.1.21vrespectively.



- Setup 2 of 2: IP Address assignment to KijanirileyWS01 and KijanirileyWS02.
- These endpoints were assigned these IP Addresses: 192.168.10 and 192.168.1.12 respectively.

```
Router_Main(config)#
Router_Main(config)#interface GigabitEthernet0/0/0
Router_Main(config-if)#ip address 192.168.1.1 255.255.255.0
Router_Main(config-if)#
```

- IP Address assignment to Router_Manufacturing.
- It was given the IP Address 192.168.1.1, and was renamed ROUTER_MAIN in the CLI.

```
SWITCH_MAIN>enable
SWITCH_MAIN#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SWITCH_MAIN(config)#hostname SWITCH_MAIN
SWITCH_MAIN(config)#
```

- Name assignment to Switch_Manufacturing.
- It was renamed to SWITCH_MAIN in the CLI.